

Synchronous vs. Asynchronous

Check out the performance difference between this face off between the two kinds of flash memory
<http://goo.gl/R4kHd>

Photonic chip in 2 years

APIC and CNSE are looking to develop a photonic microprocessor within two years.
<http://bit.ly/A7f59v>

SSD Test

capacity SSD will have more flash memory chips which can enable the controller to access more chips at a time.

Features

Intel SSDs are known to be reliable and the confidence Intel has in their SSDs is certified with a 5-year warranty whereas all the other competitors had only a 3-year warranty. The 520 series line up has SSDs ranging from 60 GB to 480 GB. We got the 240GB model for the comparison test which houses a SandForce SF 2281 controller. We were expecting an Intel based controller, but then this is not the first time Intel is using a third party controller. There are 16 x 16GB of 25nm Intel MLC NAND taking the total to 256 GB. But you will only have 240 GB at your disposal as 16GB is reserved for data parity, garbage collection and block replacement. Intel bundles in an Intel SSD Toolbox utility which lets you secure erase your

SSD from within the Windows environment. It also has an 'estimated life remaining' bar.

Coming in a flaming red metal casing, the Force GT series are the higher end SSDs in the Corsair lineup. While the 120 GB version has the SandForce SF2281 controller, the 240GB version has SF2282. The main difference being that the SF 2281 has 8 byte lanes whereas SF 2282 has 16 byte lanes, which effectively means that a single SF2282 controller can talk to twice the NAND devices as an SF2281 controller. When you open up the two SSDs, the immediate difference you will notice is the number of NAND flash. The 240GB has 32 Micron NAND chips as opposed to 16 found on the Corsair Force GT 120GB. In the 240GB drive 16GB and in the 120GB drive 8GB is kept aside for data parity, garbage collection purpose. Both the drives bundle an SSD adapter along with screws.

Kingston HyperX SSDs were among the most good looking SSDs that we have tested so far. The metal gray and blue design looks very contemporary and gives it a very premium feel. The HyperX logo is embossed on the front face. Both SSDs sport the SF2281 controller along with synchronous Intel NAND flash chips in both drives. Just like the Corsair SSDs, 16 GB and 8 GB are kept aside for data parity, garbage collection and block replacement.



Kingmax SMP35 SSD

ment in the 240 GB and 120 GB drives respectively.

It comes packed with a range of accessories. Apart from the standard SSD adapter that you get with every SSD on the market, Kingston gives you an external casing to use your SSD as an external drive - which is pointless as the USB port will be a bottleneck - but it's useful for those of you who want to carry your SSD data with you. It also bundles in a SATA cable and a screw driver to attach your SSD to your internal HDD tray.

The Kingston SSDNow V+ 200 90 GB SSD was the only SSD under 120 GB that we received for this shootout.

The V+ 200 series is an upgrade to the older, budget based SSDNow V+ 100 series of drives. It features a SandForce SF 2281 controller as opposed to the Toshiba based controller that was seen in the V+ 100 series.

HOW WE TESTED

Solid state drive testing is slightly different from your regular HDD testing. To ensure that there were absolutely no bottlenecks in the testing, we used the following test setup:

- Processor: Intel Core i7-2600K
- RAM: 2 x 4GB Kingston HyperX RAM
- Motherboard: ASRock Z68 Extreme 3 Gen 3
- OS Drive: Kingston SSDNow V+ 100 128 GB
- RAID 0 drives: 2x Kingston HyperX 120 GB drives
- Graphics Card: AMD HD 6850
- Power Supply Unit: Corsair HX 620W

We had a healthy mix of synthetic tests and real world tests. Before testing we securely erased all the SSDs using a utility called Parted Magic. You need to burn its ISO on a DVD drive and then boot into it. After which you do an Erase Disk > Internal secure erase. This ensures that the SSD is completely free of any content. This helped prevent putting the

SSDs that may have been used in the past at a disadvantage.

Crystal Disk Mark is a renowned utility known for testing drives and helps measure sequential and random read/write speeds. We kept the size at 100MB and ran 3 iterations of the test, so as to get a fair score.

AS SSD is another benchmark we used to test the sequential and random read/write scores. This test also gives us the access time which is the time required to access data from the various memory chips. We also did the Copy file benchmark test.

The final synthetic test that we ran was PC Mark 7's secondary storage test. This test stresses the SSD by simulating workloads such as adding music, importing pictures, video editing etc., to give an overall score at the end. Higher score implies a better SSD.

Synthetic tests just give you a part of the whole picture. While the scores certainly give you an idea of the transfer speeds, they are in

no way indicative of the fact that you will get the same transfer speeds in real life scenarios. We had two test files - an 8 GB sequential file and an 8 GB assorted file (which has a random mix of files ranging in capacity from a few KBs to a few GBs). These files were transferred from the RAID 0 array to the SSD file under test to simulate a write operation; from the SSD to RAID 0 array to simulate a read and finally between the two partitions of the SSD to simulate intra-drive transfer rate. The system was restarted between each of these three transfer tests, so as to get rid of any buffer.

Finally, we used Adobe Photoshop CS2 to test the amount of time taken to load a 1 GB image. Both the image and the scratch disk were kept on test drives on the same partition, effectively stimulating file access.

As solid state drives are all about performance, we did not have any weightage for features, as there are none so to speak. Ultimately everything boils down to transfer speeds and pricing.